

Curriculum Vitae

Lingyu Zhang

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Google scholar: <https://scholar.google.com/citations?user=FLZvf6MAAAAJ&hl>

EDUCATION:

Rensselaer Polytechnic Institute, Troy, NY

Ph.D. in Electrical Engineering, Aug 2021, GPA: 3.93/4.00

Columbia University, New York, NY

Master of Science in Electrical Engineering, Dec 2017, GPA: 3.71/4.00

Huazhong University of Science and Technology, Wuhan, China

Bachelor of Engineering in Optoelectronic Information Engineering, June 2016, GPA: 3.83/4.00

RESEARCH TOPIC:

Multimodal Machine Learning, Vision-Language Models, Retrieval & Recommendation Systems, Agentic AI Systems, Contextual Memory Systems, Autonomous Reasoning & Planning

WORK EXPERIENCE:

Apple, Cupertino, CA

July 2025 - Present

Generative AI Engineer - Health Sensing

- Architecting production-grade **agentic AI systems** for personalized health applications, transforming multimodal health and behavioral data into actionable user insights and adaptive recommendations.
- Building **retrieval-augmented agents** and scalable semantic retrieval systems with agentic orchestration, adaptive planning, and long-horizon reasoning capabilities.
- Developing **tool-augmented LLM systems** with contextual memory, dynamic tool usage, and multi-step reasoning across cloud and on-device models.
- Driving deployment and **inference optimization** for low-latency production AI systems under on-device and privacy constraints.
- Designing automated **evaluation harnesses** using LLM-as-a-judge methodologies to assess agent reliability, recommendation quality, and autonomous workflow performance.
- Collaborating cross-functionally with sensing, privacy, infrastructure, and product teams to deliver robust and trustworthy AI systems at scale.

Samsung Research America, Mountain View, CA

Sep 2022 - July 2025

Senior Research Scientist, Multimodal AI & Knowledge and Dialogue

AI Inventor Award, Jan 2024

- Designed and trained personalized language models for on-device question answering, enabling high-quality performance under resource constraints.
- Built efficient multimodal language models for visual question answering (VQA) and small-language-model-based planning, integrating textual and visual knowledge for robust reasoning.
- Developed compositional zero-shot learning and customized diffusion models leveraging personalized context for object recognition and image generation, advancing model adaptability and real-world applicability.

DiDi Research America, Mountain View, CA

Nov 2021 - Sep 2022

Software Engineer II in Autonomous Driving - Perception team

New Hire Award, Aug 2022

- Designed and deployed multimodal fusion models combining 2D camera and 3D LiDAR for cyclist heading estimation, improving real-world perception accuracy in safety-critical scenarios.
- Built and trained machine learning models for small object detection, leading model architecture design and optimization for high-performance perception.
- Researched and applied state-of-the-art perception algorithms, refining model training strategies and overall system design to enhance robustness across the perception stack.

INTERNSHIP EXPERIENCE:

American International Group (AIG), New York, NY September 2017 – December 2017
3D reconstruction for car accident damage estimation from single 2D image *Position: Research Assistant*

SELECTED PUBLICATIONS:

- S. Paul, **L. Zhang**, Y. Shen, and H. Jin. Enabling device control planning capabilities of small language models. 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**).
- J. S. Smith, Y. C. Hsu, **L. Zhang**, T. Hua, Z. Kira, Y. Shen, and H. Jin (2024). Continual diffusion: Continual customization of text-to-image diffusion with c-lora. Transactions on Machine Learning Research (**TMLR**).
- **L. Zhang**, T. Hua, Y. Shen, and H. Jin (2023). Prompt Tuning for Zero-shot Compositional Learning. arXiv preprint arXiv:2312.02191.
- **L. Zhang** and R.J. Radke, Natural Language Video Moment Localization Through Query-Controlled Temporal Convolution, 2022 IEEE Winter Conference on Applications of Computer Vision (**WACV**).
- **L. Zhang** and R.J. Radke, Temporal Attention and Consistency Measuring for Video Question Answering, ACM International Conference on Multimodal Interaction (**ICMI**), October 2020.
- **L. Zhang** and R.J. Radke, A Multi-Stream Recurrent Neural Network for Social Role Detection in Multiparty Interactions. **IEEE Journal** of Selected Topics in Signal Processing (2020).
- **L. Zhang**, I. Bhattacharya, M. Morgan, M. Foley, C. Riedl, B.F. Welles, and R.J. Radke, Multiparty Visual Co-Occurrences for Estimating Personality Traits in Group Meetings, 2020 IEEE Winter Conference on Applications of Computer Vision (**WACV**). IEEE, 2020.
- **L. Zhang**, M. Morgan, I. Bhattacharya, M. Foley, J. Braasch, C. Riedl, B.F. Welles, and R.J. Radke. Improved Visual Focus of Attention Estimation and Prosodic Features for Analyzing Group Interactions. ACM International Conference on Multimodal Interaction (**ICMI**), October 2019.
- M. Li, **L. Zhang**, R.J. Radke, and H. Ji, Keep Meeting Summaries on Topic: Abstractive Multi-Modal Meeting Summarization, Annual Meeting of the Association for Computational Linguistics (**ACL**), July 2019.

TEACHING EXPERIENCE

ECSE-2500 Engineering Probability
Rensselaer Polytechnic Institute, Troy, NY
ECSE-4750 Computer Graphics
Rensselaer Polytechnic Institute, Troy, NY
E6885 Reinforcement Learning
Columbia University, New York, NY

January 2019 – May 2019
Position: Teaching assistant
September 2018 – December 2018
Position: Teaching assistant
September 2017 – December 2017
Position: Course assistant

Journal Reviewer: Journal of Computer Vision and Image Understanding, Frontiers in Big Data, Information Processing Management, Transactions on Knowledge and Data Engineering

Conference Reviewer: EMNLP, ACL, WACV